

CSMVessel0000000008 Rebuilding Ships for Carrington Events

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | **Date:** June 2026 **Category:** Maritime / Synthesis

Revision Drivers (all V2.0 documents)

- 1. **2025-2026 experimental and regulatory data** supersedes V1.0 specifications
- 2. **Solar Cycle 25 SSN ~137 peak** (observed) — elevated urgency for all Carrington-focused work
- 3. **Standards updates** — AASHTO 2024, IMO 2025, IEEE P2945 draft, DNV GL EMP-2, ClassNK FRP

V1.0 vs V2.0 Parameter Changes

Parameter	V1.0	V2.0	Consequence if Not Updated
DNV EMP-2 roadmap	Not available in V1.0	Full 5-step 36-month DNV GL EMP-2 certification roadmap provided	No certification path existed in V1.0 — no bankable revenue
Total addressable market	General estimate	322 ships x \$45M avg = \$14.49B total market; 15% penetration = \$2.17B CSM	V1.0 had no quantified market sizing
Carbon intensity compliance	Not analyzed	BFRP hull 23% lighter: MEPC.80 CII improvement documented	Green shipping regulation compliance missed in V1.0
Fleet economics ROI	Not calculated	3-6x positive ROI on Aegis investment over storm expected-value modeling	V1.0 had no ROI justification for fleet owner decision
Competitive moat	Not articulated	CSM is ONLY company with complete Aegis material ecosystem for maritime	V1.0 did not position CSM as sole-source competitive advantage

Unchanged Elements

All core design philosophy, threat physics (GIC induction equations), material selection rationale, and fundamental architecture are carried forward unchanged from V1.0. Only the parameters listed above were revised.